EC991 Topics in Macroeconomics

Module Outline

University of Essex, Department of Economics
Autumn Term 2026

Module:	EC991-8-AU-CO: Topics in Macroeconomics
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Academic support hours:	Wednesdays, 13:00–15:00. Please email beforehand to confirm. Available in person and on Zoom.
Module page:	www1.essex.ac.uk/modules (EC991)
Course website:	github.com/Aseem-Patel/EC991 (materials) and Moodle (announcements, submissions).

1 Objective and Module Description

The module has two main objectives: (i) familiarise the student with cutting-edge research in macroeconomics with heterogeneous agents, and (ii) provide the necessary tools to solve these models computationally.

The module is divided into two major "blocks". In the first block, we study heterogeneity at the household level. We use models with incomplete markets where the distribution of wealth in the economy is endogenous, and individual economic shocks (income, employment, health) have strong implications for household consumption. Public policies such as taxation, unemployment insurance, and social security have first-order impacts on the well-being of families.

In the second block, we study heterogeneity at the firm level. We use models of firm and industry dynamics and entrepreneurship to study decisions related to employment, investment, innovation, market entry, and so on, and their macroeconomic implications.

Finally, we discuss how to bring these models to data: calibration, estimation, and indirect inference in macroeconomic models. The module content is adaptable, and we can dedicate more or less time to a topic depending on the interests of enrolled students.

Time permitting, we will also cover on-the-job search, wage posting and sequential auction models of the labour market, and the New Keynesian framework.

2 Assessment and Prerequisites

The module runs as follows: the lecturer presents the models and the computational methods used to solve them in the lectures, and students implement these methods in the computer classes. Assessment is by coursework. There will be two problem sets, in which students solve models on the computer using a programming language of their choice. Details of the problem sets, including deadlines, will be provided in due course.

The prerequisites are to have completed an introductory course in macroeconomics at the graduate level and to have a basic understanding of dynamic programming. Knowledge of the three-equation New Keynesian model is useful.

3 Textbooks

The module is based on papers, with core readings marked with an asterisk (*). Some textbooks are useful for specific topics in computational methods.

- Ljungqvist and Sargent (2012), *Recursive Macroeconomic Theory*: basic reference for dynamic programming. It includes a chapter dedicated to the Huggett–Aiyagari model.
- Sargent and Stachurski (2021), *QuantEcon*: open-source introduction to learning Python and Julia programming. `https://quantecon.org/`
- Heer and Maussner (2009), *Dynamic General Equilibrium Modeling*: excellent reference for methods applied to heterogeneous agent models.
- Fehr and Kindermann (2018), *Introduction to Computational Economics using Fortran*: for those interested in exploring Fortran. Useful even if you program in another language, as the algorithms are clear and well explained.

Other books are useful for different applications: Judd (1998) is the basic encyclopaedia of computational economics. Canova (2007) is excellent for DSGE and time series. Miranda and Fackler (2002) is a good reference for solutions in Matlab.

4 Module Content

1. **Household Heterogeneity**
   (a) Income and wealth inequality: the Bewley–Huggett–Aiyagari–Imrohoroglu model. Ljungqvist and Sargent (2012)*, chapters 16 and 17; Aiyagari (1994), Huggett (1993), Guvenen (2011), Heathcote et al. (2009), Achdou et al. (2022).
   (b) Consumption, income and wealth inequality over the life cycle. Storesletten et al. (2004)*, Huggett et al. (2011), Kaplan and Violante (2010).
   (c) Beyond the stationary equilibrium: transition dynamics and aggregate shocks. Auclert et al. (2021a)*, Krusell and Smith (1998)*, Boppart et al. (2018)*, Krueger et al. (2016), Algan et al. (2014).
   (d) Computational methods: endogenous grid method; Tauchen and Rouwenhorst discretisation; non-stochastic simulation of the stationary distribution; Krusell–Smith algorithm; Reiter's method. Fella (2014), Kopecky and Suen (2010), Heer and Maussner (2009), Bayer and Luetticke (2020).
   (e) Other applications (further reading): consumption and income inequality (Krueger and Perri, 2006); long-run trends in hours and income inequality (Heathcote et al., 2010); social security (Conesa and Krueger, 1999; Fuster et al., 2007); labour and capital taxation (Conesa et al., 2009; Guvenen et al., 2014); wealth inequality (De Nardi and Fella, 2017; Quadrini, 2000); fiscal policy and the wealthy hand-to-mouth (Kaplan and Violante, 2014); human capital and education (Lochner and Monge-Naranjo, 2011; Abbott et al., 2019); family economics (Greenwood et al., 2016; Barczyk and Kredler, 2018; Voena, 2015); progressive taxation (Heathcote et al., 2020; Boar and Midrigan, 2020); risk and income dynamics (De Nardi et al., 2020); consumer default (Chatterjee et al., 2007; Livshits et al., 2007); welfare and cash transfer policy (Low et al., 2010; Wellschmied, 2021); volatility shocks and consumption over the business cycle (Bayer et al., 2019; McKay, 2017); labour market frictions (Krusell et al., 2010;

Bils et al., 2011; Nakajima, 2012; Ravn and Sterk, 2021; Broer et al., 2021); fiscal policy and automatic stabilisers (Hagedorn et al., 2019; McKay and Reis, 2016); open economies (Auclert et al., 2021b); heterogeneous portfolios (Luetticke, 2021; Cui and Sterk, 2021).

2. **Firm Heterogeneity**

   (a) Firm dynamics and productivity heterogeneity. Hopenhayn (2014)*, Hopenhayn and Rogerson (1993)*, Hopenhayn (1992).

   (b) International trade. Melitz (2003).

   (c) Misallocation. Restuccia and Rogerson (2017)*, Restuccia and Rogerson (2008)*, Hsieh and Klenow (2009).

   (d) Entrepreneurship, financial frictions and development. Midrigan and Xu (2014)*, Buera et al. (2011).

   (e) Shocks, adjustment costs, and aggregate fluctuations. Clementi and Palazzo (2016)*, Khan and Thomas (2008), Bachmann and Bayer (2013).

   (f) Product and labour market power. Edmond et al. (2015)*, Berger et al. (2022), De Loecker et al. (2020).

   (g) Computational methods: projection methods; discrete choice. Terry (2017), Winberry (2018).

   (h) Other applications (further reading): innovation and quality ladders (Klette and Kortum, 2004; Atkeson and Burstein, 2019; Akcigit and Kerr, 2018); uncertainty shocks (Bloom et al., 2018; Bloom, 2009); trade liberalisation (Coşar et al., 2016; Kambourov, 2009); development and firm size (Poschke, 2018; Bento and Restuccia, 2017; Hsieh and Klenow, 2014); informality (Ulyssea, 2018; D'Erasmo and Moscoso Boedo, 2012); start-ups and firm growth (Sterk et al., 2021; Sedláček and Sterk, 2017); entrepreneurship and inequality (Allub and Erosa, 2019); size-dependent policies (Guner et al., 2008; Garicano et al., 2016); microfinance (Buera et al., 2021); large firms and granularity (Carvalho and Grassi, 2019; di Giovanni and Levchenko, 2012); firm and employment dynamics (Decker et al., 2014; Bachmann et al., 2020); customer capital (Gourio and Rudanko, 2014); manager heterogeneity (Guner et al., 2018; Akcigit et al., 2021); wealth taxation (Guvenen et al., 2019); banking industry dynamics (Corbae and D'Erasmo, 2021).

3. **Data in Macro Models: Calibration, Estimation, and Other Topics**

   (a) An (honest) discussion about calibration, estimation, and indirect inference in macro models. Nakamura and Steinsson (2018)*, Canova (2007), chapter 7.

   (b) Income and productivity processes, the wealth distribution, and MPCs. Kaplan and Violante (2022), Guvenen et al. (2021), Blundell et al. (2008).

   (c) Regional causal evidence and the missing intercept problem. Chodorow-Reich (2020), Wolf (2019), Guren et al. (2021).

4. **Additional Topics (time permitting)**

   (a) On-the-job search and wage posting. Burdett and Mortensen (1998).

   (b) Sequential auctions and individual bargaining. Postel-Vinay and Robin (2002), Cahuc et al. (2006).

   (c) The New Keynesian framework. Galí (2015), chapters 2 and 3.

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